CMPS3500

Group 5

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1. What is the size of ***LA_Crime_Data_2023_to_Present_data.csv*** and ***LA_Crime_Data_2023_to_Present_clean_data.csv***?

   LA_Crime_Data_2023_to_Present_data.csv: (390088, 21)

   LA_Crime_Data_2023_to_Present_clean_data.csv: (236813, 601)

2. What is the size of ***LA_Crime_Data_2023_to_Present_test1.csv*** and ***LA_Crime_Data_2023_to_Present_clean_test1.csv***?

   LA_Crime_Data_2023_to_Present_test1.csv: (78017, 21)

   LA_Crime_Data_2023_to_Present_clean_test1.csv: (43626, 517)

3. Describe the steps that your group took to remove duplicate rows from both data sets?

   In order to remove duplicate rows from both data sets we simply found the total count of duplicates, then wrote each duplicate to a new csv file. Upon doing this, we simply:

   df = df.drop_duplicates()

   And then checked for any lingering duplicates:

   df.duplicated().sum().

4. Describe the steps that your group took to remove outliers in all features of both data sets?

In order to remove outliers we utilized the math/data libraries given to us with Pandas and Seaborn. Before handling the outliers, we needed to detect them. We used Tukey's rule to detect these outliers. This is also known as the IQR rule. First, we calculated the interquartile range of the data (IQR = Q3 - Q1). Then we determined our outlier boundaries with the IQR. According to this rule, the data between boundaries are acceptable but the data outside of the between lower and upper boundaries are outliers. We used 1.5 times to detect IQR. We made some equations to reach outliers and indexes, and while using these we dropped around ~1,500 rows. This makes up <1% of the data.

5. Describe the steps that your group took to remove null values in all features of both data sets?

   To remove all of the null values we needed to simply drop the rows which contained no information or null values. By simply indicating null values and initiating a simple drop function we cut the row amount by around ~345 rows. Since this makes up such a small percentage of the data, there was no need to impute.

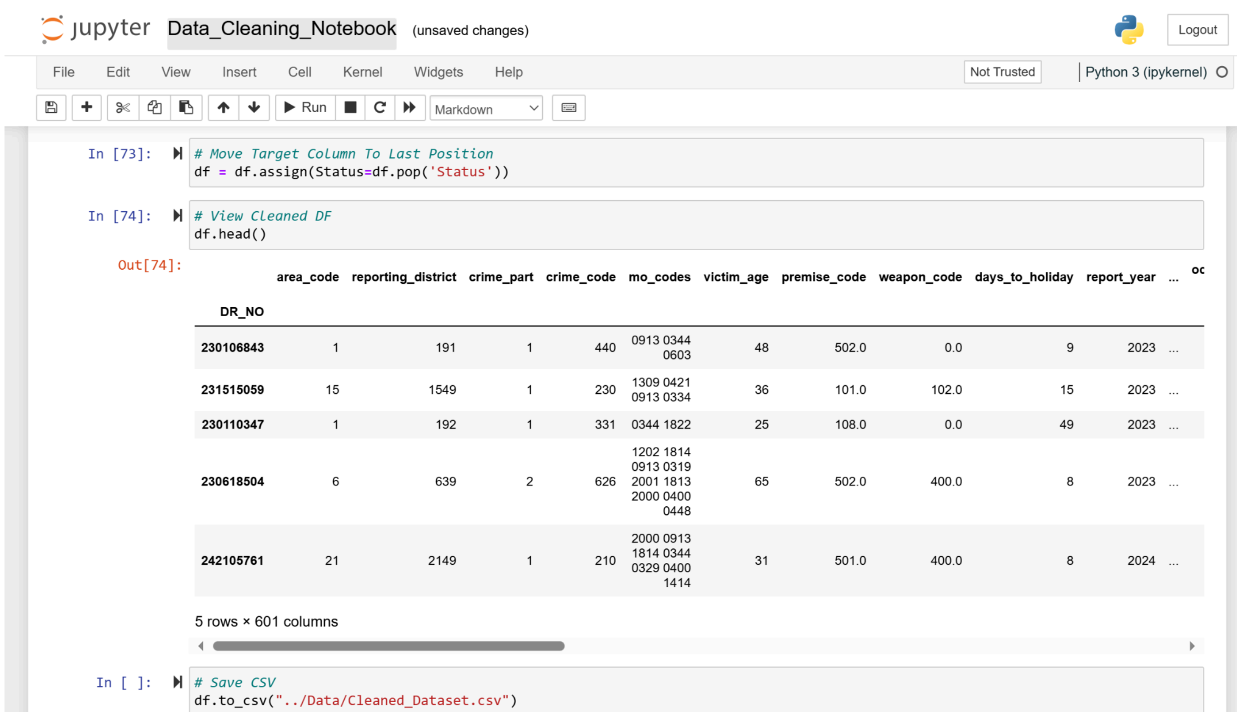
6. What rows did you drop? Explain why.

   Status, area name, date occurred, crime code description, premise description, and weapon description are the columns we dropped with their applicable rows. These rows consisted of values that were shared amongst other rows, either explicitly or transitively. These duplicates consisted of: status and status description, area name and area, date occurred and time occurred, crime code and crime code description, premise code and premise description, and lastly weapon used code and weapon description. These duplicated values are not necessary for efficient data extraction/analysis. When our group started cleaning the data frame, dropping unneeded tuples allowed us to focus on more detailed data.

7. What percentage of your code implementation has your group finished?
   (Reminder to ask in person to discuss if we are talking for the current progress of overall code implantation of the project) Personal Est, 70% assuming 30% is menus and other misc.

8. What did you group do to improve the Neural Network provided in the project? please provide an screenshot of your current version of the Neural Network of your project.

   Our Group had each individual member conduct their own version of the Neural Network, we then collaborated and discussed the strengths in each other's individual versions, and transferred them to one, the one finalized version is the Neural Network that is inline with the collaborative vision of the entire group, improving was not only to the Neural Network but to our group members as well, if there was any concepts that some group members did not understand we took the extra step to conduct discord calls to explain what each module did and why it was chosen to make sure the entire group was up to speed.



DR_NO	area_code	reporting_district	crime_part	crime_code	mo_codes	victim_age	premise_code	weapon_code	days_to_holiday	report_year	...
230106843	1	191	1	440	0913 0344 0603	48	502.0	0.0	9	2023	...
231515059	15	1549	1	230	1309 0421 0913 0334	36	101.0	102.0	15	2023	...
230110347	1	192	1	331	0344 1822	25	108.0	0.0	49	2023	...
230618504	6	639	2	626	1202 1814 0913 0319 2001 1813 2000 0400 0448	65	502.0	400.0	8	2023	...
242105761	21	2149	1	210	2000 0913 1814 0344 0329 0400 1414	31	501.0	400.0	8	2024	...

5 rows × 601 columns

9. Do you see any roadblocks that will prevent you from finishing this project?

   While no apparent roadblocks have been anticipated or visually noticed, there is no guarantee this project goes 100% roadblock free. There may be unforeseen factors that could emerge further down the road that we will have to adapt to.

10. Are all students in your group working in this project sharing the workload fairly?

    While the learning curve for data cleaning was steep for some of the group members, The more experienced members were able to patiently spearhead topics that would lead the others down the correct path for the data cleaning assignment, allowing everyone to have an equal understanding of the task at hand.

11. What is the plan of your group to make sure your software is reliable?

    We would need to implement Input Validation and Differing Data sets,

    For input validation we mean catching errors, for lets say columns that use numerical values to not be able to allow letter or error catching. On top of that we will make sure that this works with not only the data set given but with differing data sets to make sure its not just hyper adjusted to the data given.